Changes in bread consumption and 4-year changes in adiposity in Spanish subjects at high cardiovascular risk.

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The effects of bread consumption change over time on anthropometric measures have been scarcely studied. We analysed 2213 participants at high risk for CVD from the PREvención con DIeta MEDiterránea (PREDIMED) trial to assess the association between changes in the consumption of bread and weight and waist circumference gain over time. Dietary habits were assessed with validated FFQ at baseline and repeatedly every year during 4 years of follow-up. Using multivariate models to adjust for covariates, long-term weight and waist circumference changes according to quartiles of change in energy-adjusted white and whole-grain bread consumption were calculated. The present results showed that over 4 years, participants in the highest quartile of change in white bread intake gained 0·76 kg more than those in the lowest quartile (P for trend = 0·003) and 1·28 cm more than those in the lowest quartile (P for trend < 0·001). No significant dose-response relationships were observed for change in whole-bread consumption and anthropometric measures. Gaining weight (>2 kg) and gaining waist circumference (>2 cm) during follow-up was not associated with increase in bread consumption, but participants in the highest quartile of changes in white bread intake had a reduction of 33 % in the odds of losing weight (>2 kg) and a reduction of 36 % in the odds of losing waist circumference (>2 cm). The present results suggest that reducing white bread, but not whole-grain bread consumption, within a Mediterranean-style food pattern setting is associated with lower gains in weight and abdominal fat.